Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Ning, Cun-Zheng Ph.D.		
Project Title:	Technology Development of Miniaturized Far-Infrared Sources for Biomolecular Spectroscopy		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cning@mail.arc.nasa.gov	**Fax:**	FY 650-604-0987
PI Organization Type:	NASA CENTER	**Phone:**	650-604-3983
Organization Name:	NASA Ames Research Center		
PI Address 1:	M/S: 229-1		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	ATD (2004)	**Solicitation / Funding Source:**	OTHER
Start Date:	06/01/2001	**End Date:**	05/31/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Ning, Cun-Zheng		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	It has been established both theoretically and experimentally that many biomolecules such as DNA or proteins have rich spectroscopic features in the far-infrared frequency range (30~300 1/cm). Spectral features in this range provide key information about the collective motions and structures, or functions of such biomolecules. The project aims to develop a miniaturized far-infrared source based on antimonide-based semiconductor nanostructures for applications in biomolecular spectroscopy. 1) Develop an optically-pumped far-infrared(Thz) laser besed on antimonide nanostructures. 2) Develop far-infrared sources using difference-frequency-generation using two pumping lasers. 3) Demonstrate biomolecular spectroscopy using far-infrared sources.		

	Coherent and compact FIR sources are not currently easily available. The developed coherent, miniaturized Far-Infrared (FIR) source would be very important for a number of applications in biomolecular spectroscopy, in FIR astronomy, and in Earth-science-observing systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/24/2021)
Articles in Peer-reviewed Journals	Jianzhong Li, Ning CZ. "Many-body effects on intersubband resonances in narrow InAs/AlSb quantum wells." Physica E: Low-dimensional Systems and Nanostructures. 2004 Jan;20(3–4):264-7. https://doi.org/10.1016/j.physe.2003.08.015 , Jan-2004
Articles in Peer-reviewed Journals	Li J, Ning CZ. "Collective excitations in InAs quantum well intersubband transitions." Physica E: Low-dimensional Systems and Nanostructures. 2004 Apr;22(1-3):628-31. https://doi.org/10.1016/j.physe.2003.12.086 , Apr-2004
Articles in Peer-reviewed Journals	Li J, Kolokolov KL, Ning CZ, Larrabee DC, Khodaparast GA, Kono J, Ueda K, Nakajima Y, Sasa S, Inouec M. "Microscopic modeling of intersubband resonances in InAs/AlSb quantum wells." Physica E: Low-dimensional Systems and Nanostructures. 2004 Jan;20(3-4);268-71. https://doi.org/10.1016/j.physe.2003.08.016 , Jan-2004
Articles in Peer-reviewed Journals	Kolokolov K, Li J, Ning CZ. "kp Hamiltonian without spurious-state solutions." Physical Review B. 2003 Oct 15;68(16):161308. https://doi.org/10.1103/PhysRevB.68.161308 , Oct-2003
Articles in Peer-reviewed Journals	Larrabee DC, Khodaparast GA, Kono J, Ueda K, Nakajima Y, Nakai M, Sasa S, Inoue M, Kolokolov K, Li J, Ning CZ. "Temperature dependence of intersubband transitions in InAs/AlSb quantum wells." Applied Physics Letters. 2003 Nov 10;83(19):3936. https://doi.org/10.1063/1.1626264 , Nov-2003